

# Midconstruction LaHouse Meets Katrina and Rita

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**by Claudette Reichel**

**L**ouisiana House Resource Center, an educational demonstration house in Baton Rouge exhibiting four high-performance building systems, better known as LaHouse, was in midconstruction when Hurricanes Katrina and Rita ripped through Louisiana. Although it was certainly not part of the plan, this pummeling turned out to have happened at an ideal stage of construction for two reasons. First, LaHouse became an ironically timely and greatly needed exhibition of hazard-resistant, energy-efficient home-building systems, while still fully exposed to view. Second, the framing and secondary moisture barriers were put through an excellent test.

Baton Rouge received the outer bands of both storms, with approximately 50 mph sustained winds and 70–80 mph gusts, but Katrina struck from the northwest and Rita from the southeast three weeks later. At the time, LaHouse was fully framed, sheathed, and covered with house wraps and roof underlayments—but with no outer claddings.

The four LaHouse building systems include borate-treated 2 x 4 wood framing 16 inches on center (OC) with engineered-wood products; 2 x 6 wood framing spaced 24 inches OC with advanced framing (borate pressure treated); borate-treated structural insulated panels (foamcore SIPS by R-Control); and insulating concrete forms (ICF by Eco-Block) for the garage/teaching center. All systems are tied down with



This rear view of LaHouse after Katrina, shows its three types of housewrap, foil-faced OSB, ICF, and three types of roof underlayment.

hurricane connectors from Simpson Strongtie, and all systems applied the provisions recommended in the Institute of Business and Home Safety's Fortified...for Safer Living program, which stipulates building for resistance to 130 mph winds. The building systems also meet Energy Star criteria and demonstrate moisture control and mold prevention principles. The house is elevated three feet above the base flood elevation (30 inches above grade), and exhibits three types of foundation system. The garage is slab-on-grade.

## Poststorm Lessons

As expected, all four building systems held together perfectly, despite the increased pressures created by the lack of windows and doors. (When wind can enter a building, it creates internal pressures that can compound external pressures. So wind codes require stronger construction for partially enclosed buildings than for fully enclosed buildings with impact-resistant protection for glazing.) The continuous structural sheathing with blocking behind the seams, closer nailing pat-



The peel-and-stick roof membrane on the plywood decking (left side) survived Katrina, but the membrane on the unprimed OSB peeled off.

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The most enlightening finding was that the peel-and-stick roof membrane (Grace Ice and Water Shield) installed on plywood decking held tightly, but Katrina peeled off another adhesive membrane installed on oriented strand board (OSB) decking. This demonstrated the importance of closely examining the manufacturer's installation instructions. For most adhesive flashing and membrane products, the instructions recommend priming OSB for best adhesion, usually with the manufacturer's primer products. This had not been done, and the resins and finish of the OSB reduced adhesion.

So following Katrina, the OSB decking was primed with Protecto-Wrap's asphaltic primer before installing the company's Rain-Proof 60 peel-and-stick membrane on a second-story roof; it held on flawlessly through Rita.

Another pleasant surprise was the perfect condition of Dow's Styrofoam Weathermate Plus house wrap after both storms—not a single tear, and it kept the OSB skins of the SIPS dry.

The Raven Rufco-Wrap also withstood Katrina without a tear. However, Rita demonstrated the importance of making sure all window and door openings are properly cut, wrapped, and secured before a wind event. The strong southeast winds made a sail out of the wrap spanning a forgotten back doorway, which created extreme pulling pressure against its fasteners.

A key factor in preventing water damage was the window flashing products with corner guard systems that protected the sills and corners from water during severe wetting events. LaHouse used several types of window flashing systems, including Water-Out's Red Pans, Tyvek's FlexWrap, ProtectoWrap's sill drainage system, Fortifiber's flexible corner guards and adhesive flashing tape, and Raven flashing tapes. All provided effective protection.

## Resisting Mold and Corrosion

It was great to see that the borate- and copper-treated lumber and wood products used for the shell of LaHouse showed far less surface mold growth after months of exposure following the storms than did the untreated materials



High-strength synthetic roofing felt weathered both Katrina and Rita without a tear.

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terns, and metal hurricane connectors created a continuous load path and kept every framing piece securely attached, straight, and level. Nothing shifted, leaned, or pulled apart.

The most impressive outcome was the total lack of damage in the syn-

thetic roofing felt (Grace Tri-flex 30), even though it is lighter in weight than conventional #15 felt. It appeared to be the only exposed roofing felt in town that wasn't shredded after Katrina, and it didn't have a single tear after enduring both storms.



in the interior. In New Orleans and other hard-hit areas, mold has been and remains a massive problem.

Likewise, we are pleased to see that the metal connectors installed with Grace Deck Protector on copper azole-treated outside wood are still free of corrosion.

Since the storms, we at LaHouse have accelerated our plans to help disseminate critical building science information. We pause construction each Friday for Mid-Construction Open House tours and have launched a Rebuild Stronger, Safer, Smarter builder training and consumer education initiative. With support from DOE's Building America program, Building Science Corporation (BSC), and the Institute of Business and Home Safety (IBHS), Joe Lstiburek of BSC and Tim Reinhold of IBHS have developed a builder training curriculum on Best Building Practices for the Gulf Region. We couldn't be more grateful for their help and dedication.



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### **For more information:**

Louisiana House Resource Center is an educational exhibit and outreach program of the Louisiana State University AgCenter and Louisiana Cooperative Extension Service through the support of agency partners and private contributors, including Entergy Louisiana, Paula Manship, Roy O. Martin Lumber, the Louisiana Home Builders Association, a Borate-Treated Wood Alliance, and more than 80 others. For more information about LaHouse and its educational materials and programs, visit [www.LouisianaHouse.org](http://www.LouisianaHouse.org).